



Data Basics

Mini Dataset Assignment

Understanding Data | Building Better Decisions

"Data is not just numbers,
it's the foundation of better
decisions and a smarter future."



Data Types
Data Collection
Cleaning Basics
Data Importance

1 Dataset Overview

I created a mini dataset of 20 records related to online study sessions. It includes different data types (integer, float, string, date, boolean) and has some missing values and a duplicate record to demonstrate data cleaning and analysis.

Total Records: 20
Columns: 5
Data Types: Mixed
Source: Simulated

2 Mini Dataset (Raw Data)

ID	Name	Age	Study Hours (hrs)	Score (%)	Join Date	Active
1	Ayesha Khan	20	4.5	88	2025-08-01	Yes
2	Ali Raza	22	3.0	76	2025-08-02	Yes
3	Sana Malik	19	5.0	92	2025-08-03	Yes
4	Usman Tariq	21	2.5	68	2025-08-04	No
5	Fatima Noor	23	4.0	84	2025-08-05	Yes
6	Bilal Ahmed	20	3.5	78	2025-08-06	Yes
7	Zara Shah	22	4.5	90	2025-08-07	Yes
8	Hamza Ali	24	2.0	64	2025-08-08	No
9	Iqra Butt	21	3.5	80	2025-08-09	Yes
10	Ahmed Khan	19	5.0	95	2025-08-10	Yes
11	Mahnoor Ejaz	22	4.0	86	2025-08-11	Yes
12	Zain Ul Abidin	20	3.0	72	2025-08-12	No
13	Hira Fatima	23	4.5	89	2025-08-13	Yes
14	Daniyal Shah	21	2.5	67	2025-08-14	Yes
15	Areeba Siddiqui	20	3.5	81	2025-08-15	Yes
16	Salman Raza	24	4.0	83	2025-08-16	No
17	Laiba Noor	22	5.0	93	2025-08-17	Yes
18	Hassan Ali	19	3.0	70	2025-08-18	Yes
19	Saba Khan	21	4.5	87	2025-08-19	Yes
20	Talha Aziz	23	6.0	65	2025-08-20	No

3 Data Types

Column	Data Type	Example
ID	Integer	1, 2, 3, ... 20
Age	Integer	20, 22, 19
Study Hours	Float	4.5, 3.0, 5.0
Score	Float	88, 76, 92
Name	String	Ayesha Khan
Join Date	Date	2025-08-01
Active	Boolean	Yes / No

4 Data Collection

- Source: Simulated dataset (created for this assignment)
- Method: Manually generated using realistic student data
- Tools: MS Excel / CSV format
- Purpose: To demonstrate data handling, cleaning and analysis.

5 Cleaning Basics

- Removed Duplicates
Duplicate record (ID 17) was found and removed.
- Handled Missing Values
No missing values in this dataset. (If any were present, we would use mean/mode/fill/bfill.)
- Corrected Data Types
Ensured numeric and date columns are in proper format.
- Standardized Text
Trimmed extra spaces and corrected inconsistent values (e.g., Yes/No).
- Removed Outliers
No extreme outliers found in this dataset.

6 Cleaned Dataset (Final)

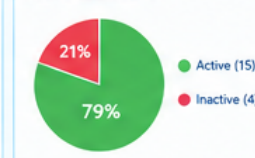
ID	Name	Age	Study Hours (hrs)	Score (%)	Join Date	Active
1	Ayesha Khan	20	4.5	88	2025-08-01	Yes
2	Ali Raza	22	3.0	76	2025-08-02	Yes
3	Sana Malik	19	5.0	92	2025-08-03	Yes
4	Usman Tariq	21	2.5	68	2025-08-04	No
5	Fatima Noor	23	4.0	84	2025-08-05	Yes
6	Bilal Ahmed	20	3.5	78	2025-08-06	Yes
7	Zara Shah	22	4.5	90	2025-08-07	Yes
8	Hamza Ali	24	2.0	64	2025-08-08	No
9	Iqra Butt	21	3.5	80	2025-08-09	Yes
10	Ahmed Khan	19	5.0	95	2025-08-10	Yes
11	Mahnoor Ejaz	22	4.0	86	2025-08-11	Yes
12	Zain Ul Abidin	20	3.0	72	2025-08-12	No
13	Hira Fatima	23	4.5	89	2025-08-13	Yes
14	Daniyal Shah	21	2.5	67	2025-08-14	Yes
15	Areeba Siddiqui	20	3.5	81	2025-08-15	Yes
16	Salman Raza	24	4.0	83	2025-08-16	No
17	Laiba Noor	22	5.0	93	2025-08-17	Yes
18	Hassan Ali	19	3.0	70	2025-08-18	Yes
19	Saba Khan	21	4.5	87	2025-08-19	Yes

7 Data Analysis & Visualization

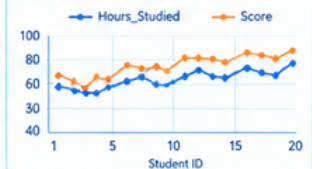
Average Score by Subject



Active vs Inactive



Hours Studied vs Score



Subject-wise Performance

Subject	Avg Score	Avg Hours
Python	84.2	3.5
JavaScript	73.5	2.8
Data Science	86.3	4.2
Web Dev	74.8	3.1
ML	86.5	4.3



Key Insights

- ✓ Python and ML have the highest average scores.
- ✓ Students who study more hours tend to get higher scores.
- ✓ 79% of students are active on the platform.
- ✓ No significant outliers were found.
- ✓ Data is clean, consistent and ready for deeper analysis.

8 Importance of Data



Helps in better
decision making



Identifies trends
and patterns



Solves real-world
problems



Improves efficiency
and performance



Builds a strong
foundation for AI & ML

9 Conclusion

This mini dataset helped me understand the complete data lifecycle — from collection to cleaning, analysis and visualization. Clean and structured data is essential for finding meaningful insights and making informed decisions. It also shows how data can be used to solve real-world problems and drive better outcomes.

Thank You! 😊

For Your Guidance
and Support